Introduction to Java

How Java code executes and more information about platform Independence... "JVM" converts byte code to machine code

.java file (human readable) $\xrightarrow{\text{Compiler}}$ (Entire file) .class file (byte code) $\xrightarrow{\text{Interpreter}}$ (line by line) Machine Code (0 & 1)

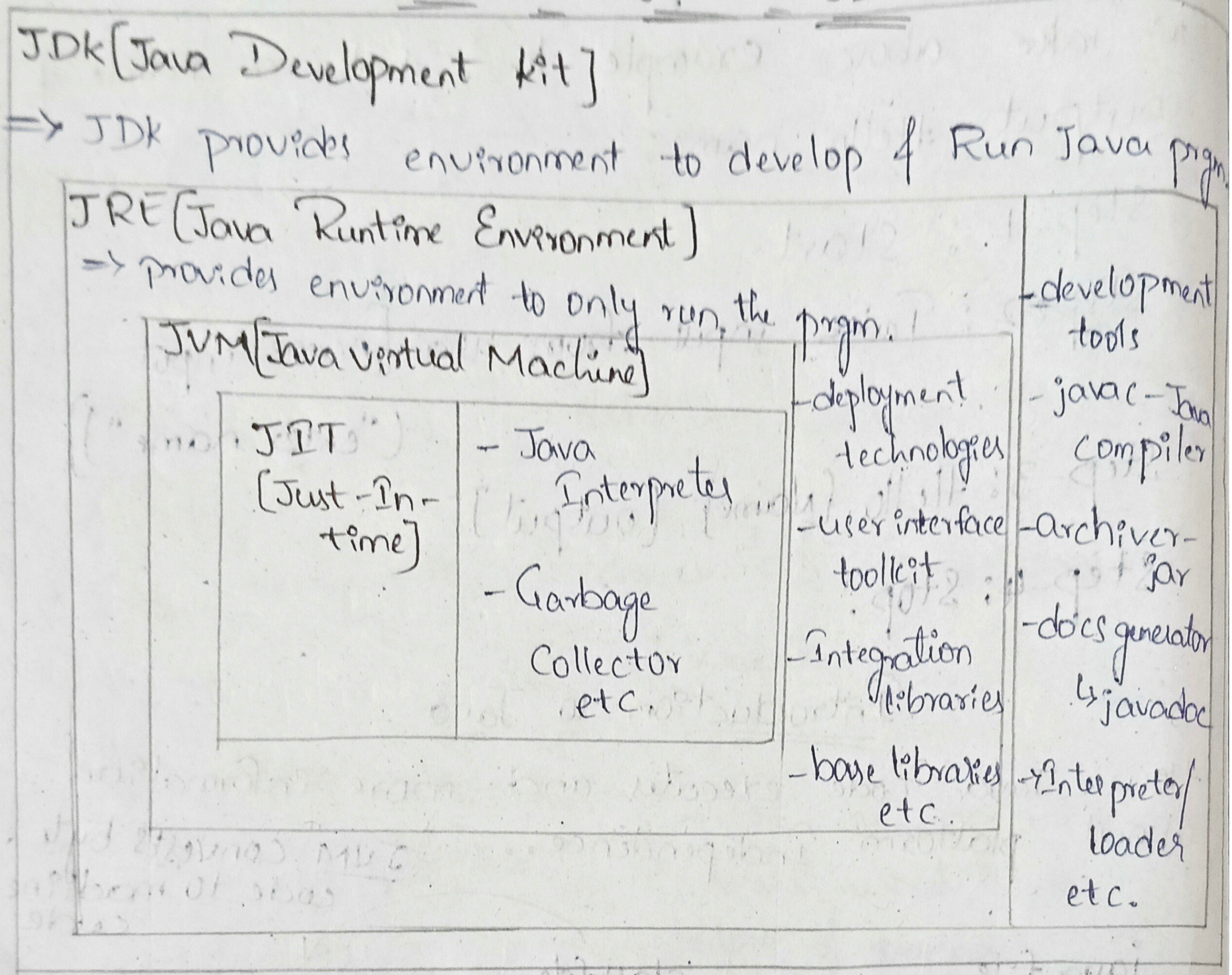
Byte code:
- It can run on all operating Systems.
- This code doesn't run directly on a System, for this we need JVM.

** Therefore, Java is platform Independent.

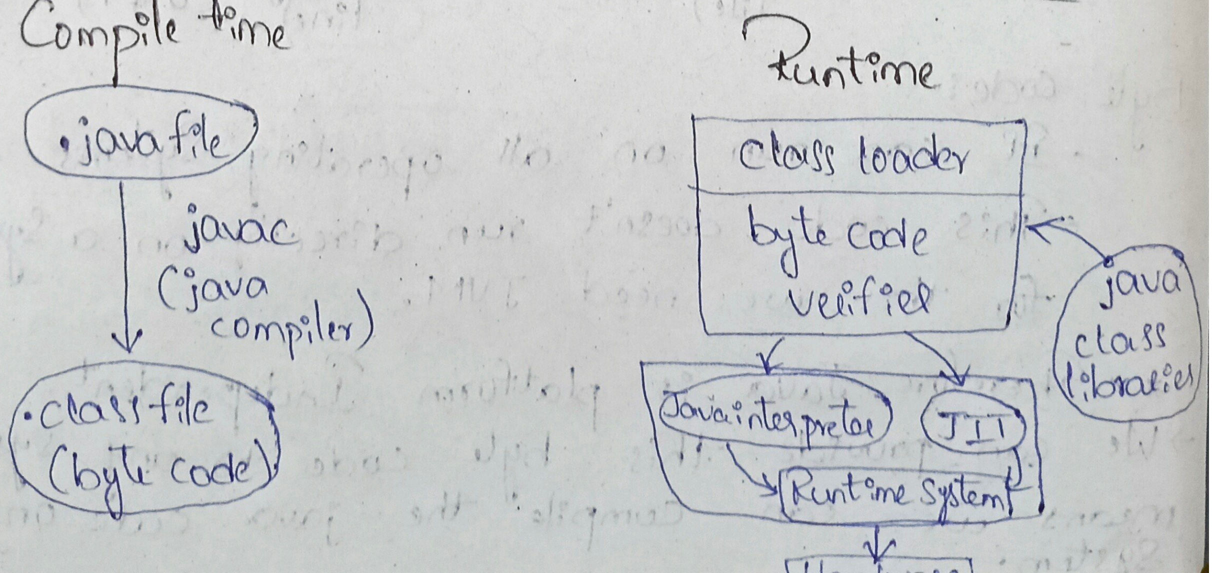
→ We can provide this byte code to any system means we can compile the java code on any System.

→ But JVM is platform dependent means, for O.S the executable file that we get, it has step by step set of ex instruction dependent on platform.

JDK vs JRE vs JVM vs JIT



Java Development and Runtime Environment



→ JVM execution:

Java Interpreter:
- line by line execution
- when one method is called many times, it will interpret again and again

JIT:
- methods that are repeated, JIT provides direct machine code so re-interpretation is not required
- makes execution faster

Garbage collector:

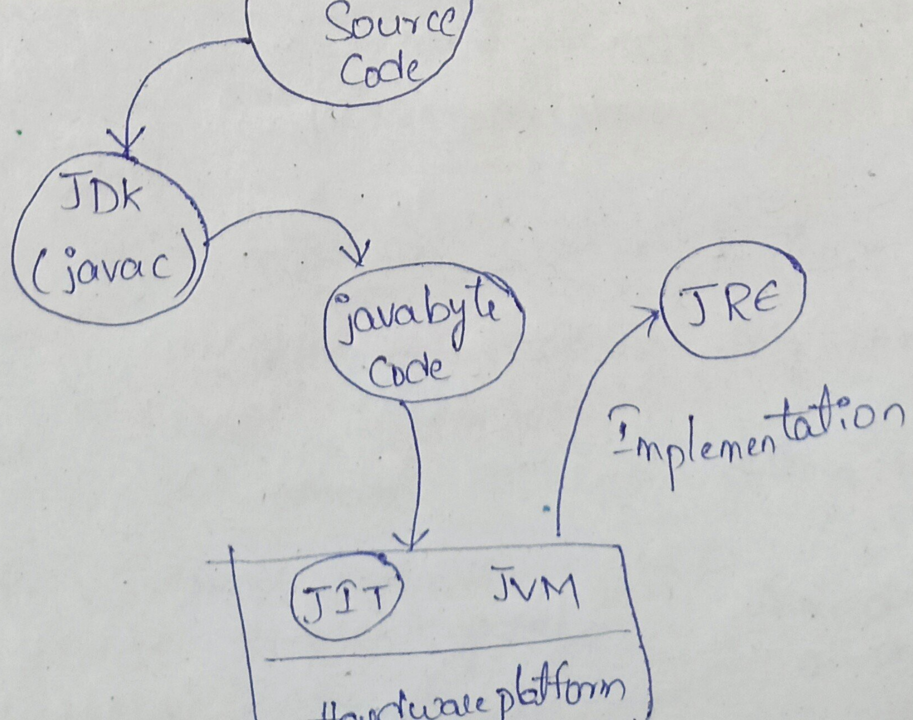
class loader:

- Loading:
- reads byte code file & generates binary data
- an object of this class is created in heap.

Linking:
- JVM verifies .class file
- allocates memory for class variables & default value.
- replace Symbolic references from the type with direct references.

Initialization:
- all static variables are assigned with their values defined in the code & static block.

Overview (Summary)



First Java Prgm - I/O, Debugging & Datatypes.

Filename: Demo.java

Class Name: Demo

→ Its good practice to use initial character as capital (you can use small also)

Public - This keyword means, it is used so that we can access the class from anywhere.

Functions - Collection of code, that we can use again & again. Functions are also known as methods.

void - The void keyword specifies that a method should not have a return value [alway].

String [ ] args: means an array of Sequence of characters ("strings") that are passed to the main function.

* After compiling, .class file is always saved in